



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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MOVIE TICKET BOOKING SYSTEM

CAPSTONE PROJECT PRESENTATION
PROGRAMMING IN PYTHON - CSA0801

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INTRODUCTION

The Movie Ticket Booking System is a menu-driven application developed using Python. It uses a MySQL database to store movie details and ticket booking information.

The system allows the user to:

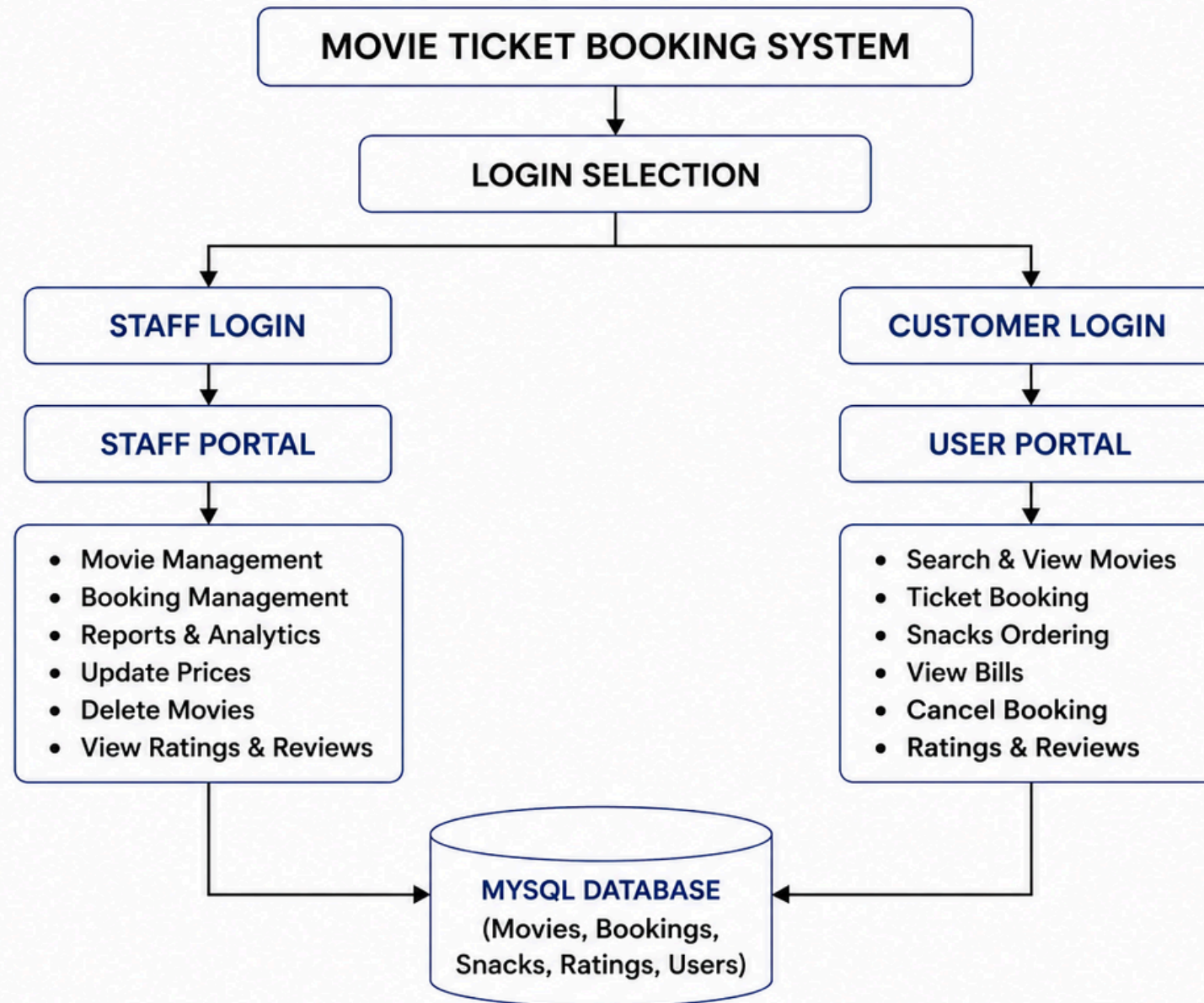
- View the list of available movies
- Book tickets for a selected movie
- Automatically calculate the total ticket amount

All data is stored securely in the database, reducing manual work and errors. This project shows how Python programs can interact with databases to perform real-time operations.

OBJECTIVES

- To manage movie details using a MySQL database.
- To search, add, update, and delete movie records.
- To book and cancel movie tickets easily.
- To manage snack orders along with movie bookings.
- To calculate movie, snack, and combined bills.
- To identify most-watched and top movies based on bookings.
- To demonstrate Python–MySQL connectivity and database operations.

SYSTEM DESIGN



METHODOLOGY

- Requirement Analysis – Identify the requirements for movie booking, movie management, snacks, ratings and billing.
- System Design – Design separate Staff and Customer portals with different functionalities.
- Database Design – Store movies, bookings, snacks and ratings in a MySQL database.
- Python Development – Implement the system using Python functions, user input and SQL queries.
- MySQL Connectivity – Connect Python with MySQL using the mysql.connector module.
- Module Development – Develop login, movie management, ticket booking, snacks, billing, ratings and reporting modules.
- Testing – Test valid inputs, invalid inputs, bookings, cancellations and database operations.
- Implementation – Integrate all modules into a complete Movie Ticket Booking System.

LITERATURE SURVEY (LATEST PAPERS)

Year	Title / Authors	Key Contribution	Pros	Cons
2024–25	A Survey on Modern Online Ticket Booking Systems (IEEE, 2025)	Studies modern online movie ticket booking systems focusing on secure QR-code ticketing, visual cryptography, and modern backend technologies.	- Improved ticket security - Reduces ticket forgery - Better data protection - Enhanced user experience	- QR codes can face security risks - Encryption adds complexity - Requires reliable digital infrastructure
2024–25	A Project on Booking Movie Online Ticket System (IEEE Access, 2024)	Provides online movie selection, show-time information, seat availability, ticket booking, snack ordering, seat modification and cancellation.	- Convenient online booking - Seat selection - Snack ordering - Cancellation support	- Depends on internet connectivity - Requires secure payment handling - System maintenance is required
2023–24	A Case Study on Online Ticket Booking System Project (IEEE, 2024)	Develops a web-based booking system with separate administrator and customer panels using HTML/CSS, PHP and JavaScript.	- Separate admin/user access - Easy navigation - Reduces booking time - Centralized management	- Web-based system requires connectivity - Security must be maintained - Backend maintenance required
2023–24	A Comparative Study in Online Movie Ticket Booking System (IEEE, 2024)	Analyzes movie listings, showtimes, seating and real-time booking updates using Node.js and MongoDB to improve speed and reliability.	- Faster processing - Real-time updates - User-friendly booking - Improved reliability	- Technology integration can be complex - Server dependency - Requires proper database management
2022–23	A Cinema – Online Movie Ticket Booking System (IEEE, 2023)	Proposes online movie selection, seat selection, payment, ticket confirmation, rescheduling, ticket transfer along with recommendation and security features.	- Multiple booking options - Ticket rescheduling/transfer - Movie recommendations - Authentication & encryption	- More complex implementation - Requires secure payment integration - Recommendation needs user data
2021–22	Online Movie Ticket Booking System (IEEE, 2021)	Presents a modular system with Theatre, User, Administrative and Transaction modules to automate movie ticket booking.	- Modular architecture - Automates booking - Supports administration - User-friendly	- Transaction management can be challenging - Requires database maintenance - Depends on system availability

MODULE 1 - REGISTRATION & LOGIN USER REGISTRATION

- Provides separate login options for Staff and Customers.
- Staff login verifies Staff ID and password.
- Customer login verifies username and password.
- Invalid staff credentials are denied access.
- Successful login redirects the user to the appropriate portal.

CODE: LOGIN MODULE

```
def staff_login():
    """
    Feature 1: Dedicated Theatre Staff Login
    """
    print("\n" + "=" * 45)
    print("          THEATRE STAFF LOGIN")
    print("=" * 45)
    username = input("Enter Staff ID / Username: ").strip()
    password = input("Enter Staff Password: ").strip()

    if ("staff" in username.lower() or "admin" in username.lower()) and password in STAFF_PASSWORDS:
        print(f"\n>>> Staff Login Successful! Welcome [{username}] <<<")
        return username
    else:
        print("\n>>> [ERROR] Invalid Staff ID or Password! Access Denied. <<<")
        return None

def user_login():
    """
    Feature 2: Dedicated User / Customer Login
    """
    print("\n" + "=" * 45)
    print("          USER / CUSTOMER LOGIN")
    print("=" * 45)
    username = input("Enter Your Username / Name: ").strip()
    password = input("Enter Password: ").strip()

    if not username or not password:
        print("\n>>> [ERROR] Username and password cannot be empty! <<<")
        return None

    print(f"\n>>> Login Successful! Welcome Customer [{username}] <<<")
    return username
```


OUTPUT: LOGIN MODULE

```
=====
MOVIE TICKET BOOKING SYSTEM
=====
1. Theatre Staff Login
2. User / Customer Login
3. Exit Program
=====
Enter your choice (1-3): 1

=====
THEATRE STAFF LOGIN
=====
Enter Staff ID / Username: admin123
Enter Staff Password: 1234

>>> Staff Login Successful! Welcome [admin123] <<<
```

MODULE 2: MOVIE MANAGEMENT

- Display all available movies.
- Add new movies to the database.
- Search movies by ID or name.
- Search movies by language.
- Search movies by price range.
- Update movie ticket prices.
- Delete movies from the database.
- All movie records are managed using MySQL.

CODE: MOVIE MANAGEMENT

```
# Function to display movies
def show_movies():
    cur.execute("SELECT * FROM movie")
    print("\n" + "=" * 40)
    print("          MOVIE LIST")
    print("=" * 40)
    rows = cur.fetchall()
    if rows:
        print("Movie ID | Title | Show Time | Price | Date")
        print("-" * 40)
        for i in rows:
            print(i)
    else:
        print("No movies available in database.")
    print("=" * 40)
```



OUTPUT: MOVIE MANAGEMENT



=====

USER PORTAL (Aaarathana)

=====

1. Show Movies
2. Search Movie (ID / Name / Language)
3. View Seat Details & Layout
4. View Theatres & Screens
5. Book Ticket (Theatre + Screen + Language + Seat)
6. Cancel Booking
7. Rate a Movie & Give Review
8. View Movie Ratings & Reviews
9. Order Snacks
10. Show Snack Bill
11. Show Total Snack Amount
12. Show Most Ordered Snack
13. Combined Bill (Movie + Snacks)
14. Logout to Main Menu

=====

Enter choice (1-14): 1

=====

MOVIE LIST

=====

Movie ID | Title | Show Time | Price | Date

```
(2, 'Coolie', '1:00 PM', 100, datetime.date(2026, 8, 25), None, None, None)
(3, 'Dragon', '4:00 PM', 100, datetime.date(2026, 8, 26), None, None, None)
(4, 'Retro', '7:00 PM', 220, datetime.date(2026, 8, 26), None, None, None)
(5, 'Interstellar', '9:30 PM', 300, datetime.date(2026, 8, 27), None, None, None)
```

=====

RESULTS & DISCUSSION

- Python successfully connected with MySQL.
- Movie records were successfully displayed and managed.
- Ticket bookings were successfully stored.
- Booking cancellation was implemented.
- Snack orders and bills were successfully calculated.
- Ratings and reviews were stored in the database.
- Most-watched movie and top movie reports were generated.
- Staff and Customer portals worked successfully.

CHALLENGES & LIMITATIONS

- Real-time seat availability is not maintained in the database.
- Staff passwords are predefined in the Python source code.
- Customer authentication is basic.
- The system is console-based.
- Online payment integration is not implemented.
- Theatre and screen information is predefined in the Python program.

FUTURE SCOPE

- Add real-time seat availability.
- Add online payment integration.
- Add email/SMS booking confirmation.
- Add administrator role management.
- Add dedicated theatre, screen and seat tables.
- Add movie posters and trailers.
- Add advanced revenue and sales reports.

CONCLUSION

- Developed an efficient Movie Ticket Booking System using Python and MySQL.
- Simplifies movie searching, ticket booking, and cancellation.
- Provides easy management of movie details and ticket prices.
- Allows users to order snacks and calculate snack costs.
- Generates a combined bill for movie tickets and snacks.
- Helps analyze booking data to identify the most-watched movies.
- Reduces manual work and improves accuracy in managing booking records.
- Demonstrates the practical use of Python–MySQL connectivity and database operations.

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Engineer to Excel



THANK YOU